

Question 1

Marks

- | | | | |
|----|-------|--|----------|
| a) | (i) | Find $\int x e^{x^2} dx$. | 1 |
| | (ii) | Find $\int \frac{x+1}{x+2} dx$. | 2 |
| b) | | Use Simpson's rule with 3 function values to estimate $\int_1^3 x \ln x \, dx$ correct to 2 decimal places. | 2 |
| c) | (i) | Express $\sqrt{3} \sin x - \cos x$ in the form of $R \sin(x - \alpha)$ where $R > 0$ and $0 \leq \alpha \leq 2\pi$. | 2 |
| | (ii) | Find the minimum of $\sqrt{3} \sin x - \cos x$. | 1 |
| | (iii) | Find the first positive value of x for which $\sqrt{3} \sin x - \cos x$ is a minimum. | 1 |

Question 2 (Start a new page)

- | | | |
|----|--|----------|
| a) | The points $A(6,5)$, $B(2,0)$ and $C(8,3)$ are the vertices of a triangle. Calculate the length of the altitude through A . | 3 |
| b) | Find $\lim_{x \rightarrow 0} (\sin 3x \div \tan \frac{x}{3})$. | 3 |
| c) | Find the area enclosed by the curve of $y = \frac{1}{\sqrt{16-x^2}}$, the x -axis and $x = 2$, $x = -2$. | 3 |

Question 3 (Start a new page)

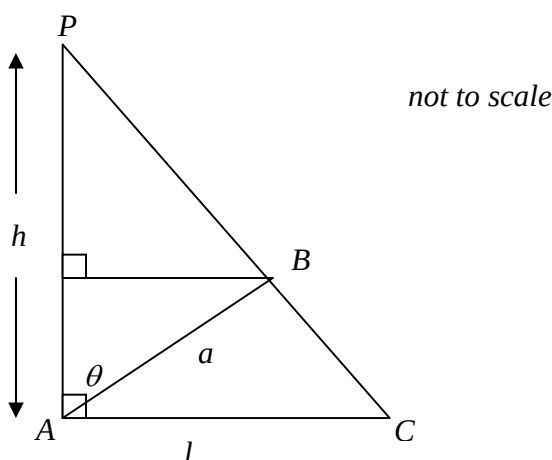
- | | | |
|----|--|----------|
| a) | Find $\int \sqrt{3x-1} \, dx$. | 1 |
| b) | Consider the function $f(x) = \sqrt{4-\sqrt{x}}$. | |
| | (i) Explain why the domain of $f(x)$ is $0 \leq x \leq 16$. | 1 |
| | (ii) Prove that $f(x)$ is a decreasing function. | 1 |
| | (iii) Find the range of $f(x)$. | 1 |
| | (iv) Find the equation of $f^{-1}(x)$. | 2 |
| | (v) By considering the graphs of $f(x)$ and $f^{-1}(x)$, prove that | 3 |
| | $\int_0^{16} \sqrt{4-\sqrt{x}} \, dx = 17 \frac{1}{15}.$ | |

Question 4 (Start a new page)**Marks**

- a) Differentiate $y = \cos^{-1}(x^2)$. 1
- b) Given that $\sin^{-1} x$ and $\cos^{-1} x$ are acute,
- (i) Show that $\sin(\sin^{-1} x - \cos^{-1} x) = 2x^2 - 1$. 2
- (ii) Solve the equation $\sin^{-1} x - \cos^{-1} x = \sin^{-1}(0.5)$. 2
- c) The region bounded by the curve $xy = 3$, the lines $x = 1$ and $y = 6$, and the axes is rotated about the y -axis. 4
Find the volume of the solid formed.

Question 5 (Start a new page)

- a) Use the substitution $u = e^x$ to find $\int e^{e^x+x} dx$. 2
- b) A rod AB of length a is hinged to a horizontal table at A . The rod is inclined to the vertical at an angle θ . There is a light located at point P at a height h vertically above A . AC of length l is the shadow of the rod on the table.



- (i) Prove that $l = \frac{ah \sin \theta}{h - a \cos \theta}$. (You may assume similar triangles without proof) 2
- (ii) Prove that as θ varies, the maximum length of the shadow is $\frac{ah}{\sqrt{h^2 - a^2}}$. 5

Question 6 (Start a new page)**Marks**

- a) Use the substitution $u = \sqrt{x}$ to evaluate $\int_1^3 \frac{dx}{\sqrt{x} + x\sqrt{x}}$. **3**
- b) (i) Sketch the graph $y = 2 \cos^{-1}\left(\frac{x}{3}\right)$. **2**
- (ii) Find the area of the region enclosed by between the graph $y = 2 \cos^{-1}\left(\frac{x}{3}\right)$, the axes and $x = \frac{3}{2}$. **4**

Question 7 (Start a new page)

- a) Find the exact value of $\tan^{-1}\left(\tan \frac{3\pi}{4}\right)$. **1**
- b) Ming was asked to sketch the graph of $y = \frac{x}{\ln(x^2)}$.
- (i) He was thinking of changing $\ln(x^2)$ to $2 \ln(x)$. Does this change the graph? Why? **1**
- (ii) Prove that $y = \frac{x}{\ln(x^2)}$ is an odd function. **1**
- (iii) Find the turning point(s) of $y = \frac{x}{\ln(x^2)}$ and determine the nature. **4**
- (iv) Sketch the graph of $y = \frac{x}{\ln(x^2)}$, showing all the important features. **2**

End of Paper

Q1.

a) $\frac{1}{2} e^{x^2} + c$ #

ii) $\int 1 - \frac{1}{x+2} dx = x - \ln(x+2) + c$ #

b) $\frac{3^{-1}}{6} (1 \ln 1 + 4 \cdot 2 \ln 2 + 3 \ln 3)$ #

$= \frac{1}{3} (8 \ln 2 + 3 \ln 3)$

$= 2.947 \dots$

$= 2.95$ (2dp) #

c) $\sqrt{3} \sin x - \cos x = R \sin x \cos \alpha - R \cos x \sin \alpha$

$\sqrt{3} = R \cos \alpha \quad 1 = R \sin \alpha$

$R = \sqrt{3+1} = 2 \quad (R > 0)$ #

$\cos \alpha = \sqrt{3}/2$ and $\sin \alpha = 1/2$

α is in Quad I

$\therefore \alpha = \frac{\pi}{6}$ #

$2 \sin(x - \frac{\pi}{6}) = \sqrt{3} \sin x - \cos x$

ii) $\min = -2$

iii) $\sin(x - \frac{\pi}{6}) = -1 = \sin(\frac{3\pi}{2})$

$x - \frac{\pi}{6} = \frac{3\pi}{2}$

$x = \frac{10\pi}{6} = \frac{5\pi}{3}$ #

Q2

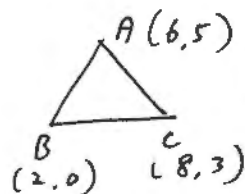
a) Equation of BC:

$\frac{y-0}{x-2} = \frac{3-0}{8-2} = \frac{1}{2}$

$y = \frac{1}{2}(x-2)$

$2y - x + 2 = 0$

altitude through A: $\left| \frac{6(-1) + 5(2) + 2}{\sqrt{5}} \right|$
 $= \frac{6}{\sqrt{5}}$ #



2b) $\lim_{x \rightarrow 0} \left(\frac{\sin 3x}{3x} \right)^{\left(\frac{x}{\tan x} \right)} = 9$ #

$= 1 \times 1 \times 9$ #

$= 9$ #

2c) Area = $\int_{-2}^2 \frac{dx}{\sqrt{16-x^2}}$ #

$= 2 \int_0^2 \frac{dx}{\sqrt{16-x^2}}$ (Even function)

$= 2 \left[\sin^{-1}\left(\frac{x}{4}\right) \right]_0^2$ #

$= 2 \left[\sin^{-1}\left(\frac{1}{2}\right) - 0 \right]$

$> \frac{\pi}{3}$ #

Q3

a) $\frac{2}{3} (3x-1)^{3/2} + c$

$= \frac{2}{9} (3x-1)^{3/2} + c$ #

b) $\sqrt{4-\sqrt{x}} \geq 0$

$4 - \sqrt{x} \geq 0$

$4 \geq \sqrt{x}, x \geq 0$

$\therefore 16 \geq x \geq 0$ #

ii) $f'(x) = \frac{1}{2\sqrt{4-\sqrt{x}}} \cdot \frac{-1}{2\sqrt{x}}$

$f'(x) = \frac{-1}{4\sqrt{x}\sqrt{4-\sqrt{x}}} < 0$ #

$\therefore f(x)$ is a decreasing function #

(iii) $f(x) = \sqrt{4 - \sqrt{x}}$

$0 \leq x \leq 16$

when $x=0$, $f(x)=2$

$x=16$ $f(x)=0$

but $f(x) \geq 0$

$\therefore 0 \leq f(x) \leq 2$

iv) Given $y = \sqrt{4 - \sqrt{x}}$ $0 \leq x \leq 16$
 $0 \leq y \leq 2$

To find $f^{-1}(x)$:

$x = \sqrt{4 - \sqrt{y}}$

$x^2 = 4 - \sqrt{y}$

$\sqrt{y} = 4 - x^2$

$f^{-1}(x) = y = (4 - x^2)^2, 0 \leq x \leq 2$

v) $\int_0^{16} \sqrt{4 - \sqrt{x}} dx = \int_0^2 (4 - x^2)^2 dx$

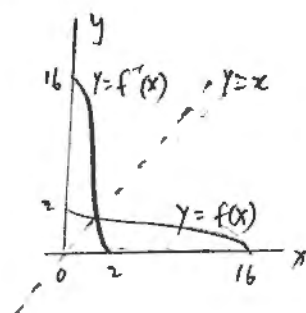
$= \int_0^2 (16 - 8x^2 + x^4) dx$

$= \left[16x - \frac{8}{3}x^3 + \frac{x^5}{5} \right]_0^2$

$= 32 - \frac{64}{3} + \frac{32}{5}$

$= \frac{480 - 320 + 96}{15}$

$= \frac{256}{15}$
 $= 17 \frac{1}{15}$

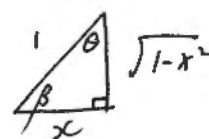


u 4

v2

a) $y' = \frac{-1}{\sqrt{1-x^4}} \cdot 2x = \frac{-2x}{\sqrt{1-x^4}}$

b(i) Let $\theta = \sin^{-1} x$
 $\beta = \cos^{-1} x$



$\sin(\sin^{-1} x - \cos^{-1} x)$

$= \sin(\theta - \beta) = \sin \theta \cos \beta - \cos \theta \sin \beta$

$= x \cdot x - (\sqrt{1-x^2})(\sqrt{1-x^2})$

$= x^2 - (1 - x^2)$

$= 2x^2 - 1$

ii) Solve $\sin^{-1} x - \cos^{-1} x = \sin^{-1}(0.5)$

$\sin(\sin^{-1} x - \cos^{-1} x) = \sin(\sin^{-1}(0.5))$

Fr. (i) $2x^2 - 1 = 0.5$

$2x^2 = 1.5$

$x^2 = 0.75 = \frac{3}{4}$

$x = \pm \frac{\sqrt{3}}{2}$

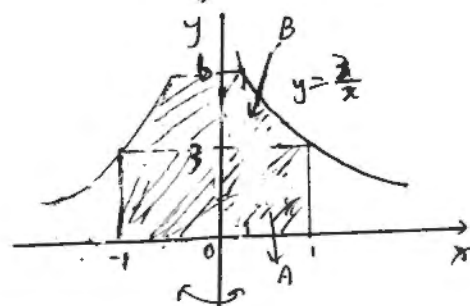
but θ, β are acute

$1 \geq x \geq 0$

$\therefore x = \frac{\sqrt{3}}{2}$ only

c)

$x=1, y=3$



Vol(A) = vol of cylinder

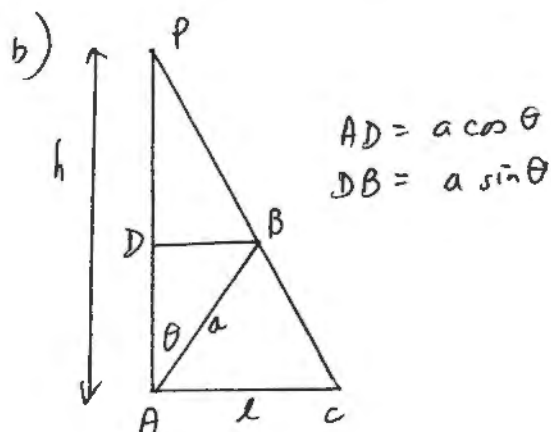
$= (\pi \cdot 1^2) 3 = 3\pi$

Vol B = $\int_3^6 \pi x^2 dy = \pi \int_3^6 \left(\frac{3}{y}\right)^2 dy$

$= 9\pi \int_3^6 \frac{dy}{y^2} = 9\pi \left[-\frac{1}{y} \right]_3^6 = -\frac{9\pi}{6} + \frac{9\pi}{3} = \frac{3\pi}{2}$

Total vol = $3\pi + \frac{3\pi}{2} = \frac{9\pi}{2}$ unit³

a) $u = e^x \quad du = e^x dx$
 $\int e^x \cdot e^x dx = \int e^u du$
 $= e^u + c = e^{e^x} + c$ #



(i) $\triangle PBD \sim \triangle PAC$

$\therefore \frac{DB}{AC} = \frac{PD}{PA}$

$\frac{a \sin \theta}{AC} = \frac{h - a \cos \theta}{h}$

$\frac{a \sin \theta}{l} = \frac{h - a \cos \theta}{h}$

$l = \frac{ah \sin \theta}{h - a \cos \theta}$ #

(ii) $l = \frac{ah \sin \theta}{h - a \cos \theta}$

$\frac{dl}{d\theta} = \frac{(h - a \cos \theta)ah \cos \theta - (ah \sin \theta)a \sin \theta}{(h - a \cos \theta)^2}$

$\frac{dl}{d\theta} = 0$
 $ah^2 \cos \theta - a^2 h \cos^2 \theta - a^2 h \sin^2 \theta = 0$
 $ah^2 \cos \theta - a^2 h = 0$

$h \cos \theta = a$
 $\cos \theta = \frac{a}{h}$ #

Since $\cos \theta$ is a decreasing function for $0 \leq \theta \leq \frac{\pi}{2}$ and limit $\varepsilon \rightarrow 0^+$

$\cos(\theta + \varepsilon) > \frac{a}{h}$

$\cos(\theta - \varepsilon) < \frac{a}{h}$

θ	$\cos^{-1}\left(\frac{a}{h}\right) - \varepsilon$	$\cos^{-1}\left(\frac{a}{h}\right)$	$\cos^{-1}\left(\frac{a}{h}\right) + \varepsilon$
$\frac{dl}{d\theta}$	+	0	-

$\therefore \theta = \cos^{-1}\left(\frac{a}{h}\right)$ is rel max

Since l is continuous for $0 \leq \theta \leq \frac{\pi}{2}$ and no other T.P

$\theta = \cos^{-1}\left(\frac{a}{h}\right)$ give absolute max $\frac{1}{2}$

$\sin \theta = \frac{\sqrt{h^2 - a^2}}{h}$
 $\cos \theta = \frac{a}{h}$

$l = \frac{ah \sin \theta}{h - a \cos \theta} = \frac{a \sqrt{h^2 - a^2}}{h - \frac{a^2}{h}}$

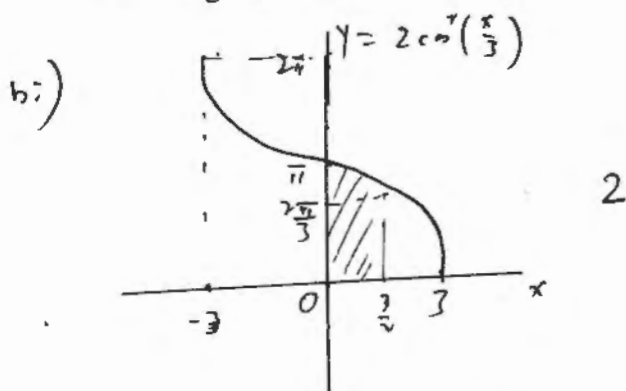
$l = \frac{a \sqrt{h^2 - a^2}}{h - \frac{a^2}{h}} \cdot h = \frac{ah}{\sqrt{h^2 - a^2}}$ #

Q6

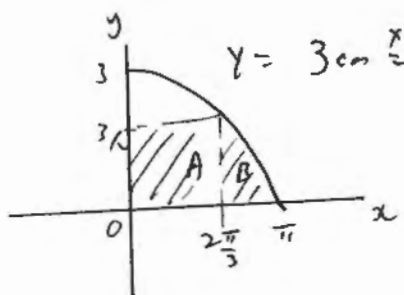
a) $u = \sqrt{x}$
 $du = \frac{dx}{2\sqrt{x}}$
 $2u du = dx$
 $x=1, u=1$
 $x=3, u=\sqrt{3}$

$$\int_1^{\sqrt{3}} \frac{2u du}{u(1+u^2)} = 2 \int_1^{\sqrt{3}} \frac{du}{1+u^2} = 2 \left[\tan^{-1} u \right]_1^{\sqrt{3}}$$

$$= 2 \left[\frac{\pi}{3} - \frac{\pi}{4} \right] = \frac{\pi}{2}$$



$$x = \frac{\pi}{2}, y = 2 \cos^2\left(\frac{\pi}{6}\right) = \frac{3}{2}$$



Part A is a rectangle

$$\text{Area A} = \frac{2\pi}{3} \cdot \frac{3}{2} = \pi$$

$$\text{Area Part B} = \int_{\frac{2\pi}{3}}^{\pi} 3 \cos^2 \frac{x}{3} dx$$

$$= 3 \cdot 2 \left(\sin \frac{x}{3} \right)_{\frac{2\pi}{3}}^{\pi}$$

$$= 6 - 6\sqrt{3} = 6 - 3\sqrt{3}$$

$$\text{Total (A+B)} = \pi + 6 - 3\sqrt{3} \text{ units}^2$$

Q7

a) $\frac{\pi}{4}$

b) Yes, lost left half of the graph.

For $y = \frac{x}{\ln x}$ D: $x \in \mathbb{R}$ but $x \neq 0, \pm 1$

For $y = \frac{x}{2\ln x}$ D: $x > 0$ but $x \neq 1$

ii) $f(-x) = \frac{-x}{\ln(-x)} = \frac{-x}{\ln x} = -f(x)$
 $\therefore$ odd

iii) Since $f(x)$ is odd, only need to consider values of $x > 0$ and use point symmetry.

$$y = \frac{x}{\ln x} = \frac{x}{2\ln x} \quad x > 0, x \neq 1$$

$$y' = \frac{2\ln x - x \cdot \frac{2}{x}}{(2\ln x)^2} = \frac{2(\ln x - 1)}{4(\ln x)^2}$$

$$y' = \frac{\ln x - 1}{2(\ln x)^2}$$

SP $y' = 0$ when $x = e, y = \frac{e}{2}$

check max/min

x	2	e	3
y'	-0.32	0	0.04

local min at $(e, \frac{e}{2})$

Using point symmetry local max at $(-e, -\frac{e}{2})$

